Homework Assignment 6

Recommended completion: Thursday, November 5, 2026, 11:59 p.m. Mountain time
Hard deadline: Tuesday, November 17, 2026, 11:59 p.m. Mountain time
Total marks: 26

Policies:

For all multiple-choice questions, select one answer.
This PDF version of the questions has been provided for your convenience should you wish to print them and work offline.
Only answers submitted through the Canvas quiz system will be graded. Please do not submit a written copy of your responses.
Completion-based grading. Submit a complete assignment by the hard deadline to receive full marks. Canvas may display question-level scores to show how you did, but those scores do not reduce your assignment mark.

Question 1. [1 MARK]

Consider the predictor $f(x) = xw$, where $w \in \mathbb{R}$ is a one-dimensional parameter, and x represents the feature with no bias term. Suppose you are given a dataset of n data points $\mathcal{D} = ((x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n))$, where each y_i is the target variable corresponding to feature x_i. Let the loss function be the scaled squared loss $\ell(f(x), y) = c(f(x) - y)^2$ where $c > 0$. Assume $n \geq 1$ and $\sum_{i=1}^n x_i^2 > 0$. The estimate of the expected loss for a parameter $w \in \mathbb{R}$ is defined as the following convex function:

$$\hat{L}(w) = \frac{1}{n} \sum_{i=1}^{n} c(x_i w - y_i)^2$$

What is the closed form solution for $\hat{w} = \arg\min_{w \in \mathbb{R}} \hat{L}(w)$?

a. $\hat{w} = \frac{\sum_{i=1}^n c x_i y_i}{\sum_{i=1}^n x_i^2}$

b. $\hat{w} = \frac{\sum_{i=1}^n y_i}{n}$

c. $\hat{w} = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i}$

d. $\hat{w} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$

Question 2. [1 MARK]

Let everything be defined as in the previous question. Suppose we consider the multivariate case where $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{w}$, and $\mathbf{w} \in \mathbb{R}^{d+1}$. What is the closed form solution for $\hat{\mathbf{w}} = \arg\min_{\mathbf{w} \in \mathbb{R}^{d+1}} \hat{L}(\mathbf{w})$?

a. $\hat{\mathbf{w}} = A^{-1} b$ where $A = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top$ and $b = \sum_{i=1}^n \mathbf{x}_i y_i$ (assume that A is invertible).

b. $\hat{\mathbf{w}} = Ab$ where $A = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top$

c. $\hat{\mathbf{w}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$

d. $\hat{\mathbf{w}} = cA^{-1} b$

Question 3. [1 MARK]

Let $g(w) = -\ln w \sum_{i=1}^{n} y_i - \ln(1-w) \sum_{i=1}^{n} (1-y_i)$ where $0 < w < 1$ and $y_i \in \{0, 1\}$. We can rewrite this a bit more simply as $g(w) = -s \ln w - (n-s) \ln(1-w)$, where $s = \sum_{i=1}^{n} y_i$. Assume $0 < s < n$. What is the derivative $g'(w)$ and the first order gradient descent update rule with a constant step size η?

a. $g'(w) = -\frac{s}{1-w} + \frac{n-s}{w}$ and update rule $w \leftarrow w - \eta \left(-\frac{s}{1-w} + \frac{n-s}{w}\right)$

b. $g'(w) = -\frac{s}{w} + \frac{n-s}{1-w}$ and update rule $w \leftarrow w - \eta \left(-\frac{s}{1-w} + \frac{n-s}{w}\right)$

c. $g'(w) = -\frac{s}{w} + \frac{n-s}{1-w}$ and update rule $w \leftarrow w - \eta \left(-\frac{s}{w} + \frac{n-s}{1-w}\right)$

d. $g'(w) = -\frac{s}{1-w} - \frac{n-s}{w}$ and update rule $w \leftarrow w - \eta \left(-\frac{s}{1-w} - \frac{n-s}{w}\right)$

Question 4. [1 MARK]

Let everything be defined as in the previous question. What is the second derivative $g''(w)$ and the Newton update rule (with unit step size)?

a. $g''(w) = \frac{s}{w^2} - \frac{n-s}{(1-w)^2}$ and update: $w \leftarrow w - \frac{-\frac{s}{w} + \frac{n-s}{1-w}}{\frac{s}{w^2} - \frac{n-s}{(1-w)^2}}$

b. $g''(w) = \frac{s}{w^2} + \frac{n-s}{(1-w)^2}$ and update: $w \leftarrow w - \frac{-\frac{s}{w} + \frac{n-s}{1-w}}{\frac{s}{w^2} + \frac{n-s}{(1-w)^2}}$

c. $g''(w) = -\frac{s}{w^2} + \frac{n-s}{(1-w)^2}$ and update: $w \leftarrow w - \frac{-\frac{s}{w} + \frac{n-s}{1-w}}{-\frac{s}{w^2} + \frac{n-s}{(1-w)^2}}$

d. $g''(w) = \frac{s}{w^2} + \frac{n-s}{(1-w)^2}$ and update: $w \leftarrow w + \frac{-\frac{s}{w} + \frac{n-s}{1-w}}{\frac{s}{w^2} + \frac{n-s}{(1-w)^2}}$

Question 5. [1 MARK]

Let everything be defined as in the previous question. What is the closed form solution for

$$w^* = \arg \min_{0<w<1} g(w)$$

a. $w^* = n/s$

b. $w^* = s/(n-s)$

c. $w^* = s/(s-n)$

d. $w^* = s/n$

Question 6. [1 MARK]

Let $g(w) = w^4 + e^{-w}$ where $w \in \mathbb{R}$. What is the derivative $g'(w)$ and the first order gradient descent update rule with a constant step size η?

a. $g'(w) = 4w^3 - e^{-w}$ and update: $w \leftarrow w - \eta(4w^3 - e^{-w})$

b. $g'(w) = 4w^3 + e^{-w}$ and update: $w \leftarrow w - \eta(4w^3 + e^{-w})$

c. $g'(w) = 4w^3 + e^{-w}$ and update: $w \leftarrow w + \eta(4w^3 + e^{-w})$

d. $g'(w) = 4w^3 - e^{-w}$ and update: $w \leftarrow w + \eta(4w^3 - e^{-w})$

Question 7. [1 MARK]

Let everything be defined as in the previous question. What is the second derivative $g''(w)$ and the Newton update rule (with unit step size)?

a. $g''(w) = 12w^2 - e^{-w}$ and update: $w \leftarrow w - \frac{4w^3 - e^{-w}}{12w^2 - e^{-w}}$

b. $g''(w) = 12w^2 + e^{-w}$ and update: $w \leftarrow w + \frac{4w^3 - e^{-w}}{12w^2 + e^{-w}}$

c. $g''(w) = 12w^2 + e^{-w}$ and update: $w \leftarrow w - \frac{4w^3 - e^{-w}}{12w^2 + e^{-w}}$

d. $g''(w) = 12w^2 - e^{-w}$ and update: $w \leftarrow w + \frac{4w^3 - e^{-w}}{12w^2 - e^{-w}}$

Question 8. [1 MARK]

Let everything be defined as in the previous question. For the Newton update rule, calculate $w^{(1)}$ if $w^{(0)} = 0$.

(Do not write your answer as a fraction. Instead, express it as a decimal number, rounded to two decimal places if necessary. For example, write 0.33 instead of 1/3.)

Question 9. [1 MARK]

Let everything be defined as in the previous question. Use the normalized-gradient update $w^{(t+1)} = w^{(t)} - \eta\, g\prime(w^{(t)})/(|g\prime(w^{(t)})| + \epsilon)$. For the first order update rule, calculate $w^{(1)}$ if $w^{(0)} = 0$, $\eta = 1$. Only for this problem, set $\epsilon = 0$.

(Do not write your answer as a fraction. Instead, express it as a decimal number, rounded to two decimal places if necessary. For example, write 0.33 instead of 1/3.)

Question 10. [1 MARK]

Let $g(\mathbf{w}) = g(w_1, w_2) = w_1^2 w_2^2 + e^{-w_1} + e^{-w_2}$ where $\mathbf{w} \in \mathbb{R}^2$. What is the gradient of $g(w)$ and the first order gradient descent update rule with a constant step size η?

a. $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \left(2w_1^{(t)}(w_2^{(t)})^2, 2w_2^{(t)}(w_1^{(t)})^2\right)^\top$

b. $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \left(2w_1^{(t)}(w_2^{(t)})^2 - e^{-w_1^{(t)}}, 2w_2^{(t)}(w_1^{(t)})^2 - e^{-w_2^{(t)}}\right)^\top$

c. $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \left(2w_1^{(t)}(w_2^{(t)})^2 + e^{-w_1^{(t)}}, 2w_2^{(t)}(w_1^{(t)})^2 + e^{-w_2^{(t)}}\right)^\top$

d. $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \left(-2w_1^{(t)}(w_2^{(t)})^2 + e^{-w_1^{(t)}}, -2w_2^{(t)}(w_1^{(t)})^2 + e^{-w_2^{(t)}}\right)^\top$

Question 11. [1 MARK]

Assume both minima exist. If $\mathcal{F} \subseteq \mathcal{G}$, then is it true that $\min_{f \in \mathcal{F}} \hat{L}(f) \geq \min_{g \in \mathcal{G}} \hat{L}(g)$?

Question 12. [1 MARK]

Consider the setting of polynomial regression. Let $d = 2$, such that $\mathbf{x} = (x_0 = 1, x_1, x_2)$, and $p = 4$, where $\bar{p}$ counts all monomials of total degree at most p, including the constant term. Then $\bar{p} = 10$. True or False?

Question 13. [1 MARK]

Let everything be defined as in the previous question. The expression for $\phi_p(\mathbf{x})$ is given by

$$\phi_p(\mathbf{x}) = \big(x_1, x_2, x_1^2, x_1x_2, x_2^2, \\ x_1^3, x_1^2x_2, x_1x_2^2, x_2^3, x_1^4, x_1^3x_2, x_1^2x_2^2, x_1x_2^3, x_2^4\big)\,.$$

True or False?

Question 14. [1 MARK]

Let $\phi_p(\mathbf{x})$ contain all monomials of total degree at most p, including the constant term. On a fixed input domain $\mathcal{X} \subseteq \mathbb{R}^{d+1}$, define

$$\overline{\mathcal{F}}_p = \left\{ f : \mathcal{X} \to \mathbb{R} \;\middle|\; \begin{array}{c} f(\mathbf{x}) = \log(\phi_p(\mathbf{x})^\top \mathbf{w}), \\ \phi_p(\mathbf{x})^\top \mathbf{w} > 0 \text{ for every } \mathbf{x} \in \mathcal{X} \end{array} \right\},$$

where the parameter vector has the dimension required by ϕ_p. Is $\overline{\mathcal{F}}_1 \subseteq \overline{\mathcal{F}}_2$? True or False?

Question 15. [1 MARK]

You are predicting house prices. Supose you want to make the irreducible error smaller. Under squared loss, can adding a new feature that provides predictive information about price beyond the existing features decrease the irreducible error? True or False?

Question 16. [1 MARK]

Consider the same setting as the previous problem. Reducing the number of i.i.d. training examples is generally a good way to reduce estimation error. True or False?

Question 17. [1 MARK]

Consider the same setting as the previous problem. Using a larger function class that contains the original class can reduce approximation error (and cannot increase it). True or False?

Question 18. [1 MARK]

You notice your predictor is overfitting. To reduce overfitting, we should make the degree p of the polynomial function class larger. True or False?

Question 19. [1 MARK]

Consider the usual bias–variance pattern for empirical risk minimization. Suppose you have a small dataset and a flexible function class that can closely fit the training data. Would the variance be large or small? Would you expect the bias to be large or small? Would you expect the predictor $\hat{f}_D$ to be underfitting or overfitting the data or neither?

a. variance large, bias large, overfit.

b. variance small, bias large, overfit.

c. variance large, bias small, overfit.

d. variance small, bias small, underfit.

Question 20. [1 MARK]

Consider the usual bias–variance pattern. Suppose you have a large dataset, but a small function class, and f_{Bayes} is much more complex than any function in the function class. Would the variance be large or small? Would you expect the bias to be large or small? Would you expect the predictor $\hat{f}_D$ to be underfitting or overfitting the data or neither?

a. variance large, bias large, underfit.

b. variance small, bias large, underfit.

c. variance small, bias small, neither overfitting nor underfitting.

d. variance large, bias large, neither overfitting nor underfitting.

Question 21. [1 MARK]

Consider the usual bias–variance pattern. Suppose you have a large dataset, a small function class $\mathcal{F}$, and $f_{\text{Bayes}} \in \mathcal{F}$. Would the variance be large or small? Would you expect the bias to be large or small? Would you expect the predictor $\hat{f}_D$ to be underfitting or overfitting the data or neither?

a. variance large, bias large, overfitting.

b. variance small, bias small, overfitting.

c. variance small, bias small, neither overfitting nor underfitting.

d. variance large, bias large, neither overfitting nor underfitting.

Question 22. [1 MARK]

You are using regularization. You notice you are underfitting. Decreasing the regularization strength λ is a reasonable change to try to reduce underfitting; use validation data to check whether it improves prediction. True or False?

Question 23. [1 MARK]

Suppose you have a dataset $\mathcal{D} = (z_1, \ldots, z_n)$ containing n i.i.d. flips of a coin. Since the flips are i.i.d. you know they all follow the distribution Bernoulli (α^*). However, you do not know what α^* is so you would like to estimate it using MLE. Which of the following is the maximum likelihood estimate α_{MLE}?

a. $\alpha_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} \alpha_i$

b. $\alpha_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} z_i$

c. $\alpha_{\text{MLE}} = \frac{1}{n-1} \sum_{i=1}^{n} z_i$

d. $\alpha_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n-1} z_i$

Question 24. [1 MARK]

Assume that $Y|X$ follows a Gaussian distribution with mean $\mu = xw_1$ and variance $\sigma^2 = \exp(xw_2)$ for all $x \in \mathbb{R}$ and $\mathbf{w} = (w_1, w_2)$ where $w_1, w_2 \in \mathbb{R}$. The negative log-likelihood, can be written as follows for a dataset $\mathcal{D} = ((x_1, y_1), \cdots, (x_n, y_n))$:

$$g(\mathbf{w}) = \sum_{i=1}^{n} g_i(\mathbf{w}) \quad \text{where } g_i(\mathbf{w}) = -\ln p(y_i|x_i, \mathbf{w}),$$

where $p(\cdot|\cdot)$ is the density of the above Gaussian distribution. What is partial derivative of g with respect to w_1?

a. $\frac{\partial g}{\partial w_1} = \sum_{i=1}^{n} \frac{x_i(y_i - x_i w_1)}{\exp(x_i w_2)}$

b. $\frac{\partial g}{\partial w_1} = \sum_{i=1}^{n} \frac{(y_i - x_i w_1)^2}{2\exp(x_i w_2)}$

c. $\frac{\partial g}{\partial w_1} = -\sum_{i=1}^{n} \frac{x_i(y_i - x_i w_1)}{\exp(x_i w_2)}$

d. $\frac{\partial g}{\partial w_1} = -\sum_{i=1}^{n} \frac{(y_i - x_i w_1)^2}{\exp(x_i w_2)}$

Question 25. [1 MARK]

Let everything be defined as in the previous question. What is partial derivative of g with respect to w_2?

a. $\sum_{i=1}^{n} \left(-\frac{(y_i - x_i w_1)^2}{2\exp(x_i w_2)} + x_i \right)$

b. $\sum_{i=1}^{n} \left(\frac{(y_i - x_i w_1)^2}{2\exp(x_i w_2)} + \frac{x_i}{2} \right)$

c. $\sum_{i=1}^{n} \left(\frac{x_i(y_i - x_i w_1)^2}{2\exp(x_i w_2)} - \frac{x_i}{2} \right)$

d. $\sum_{i=1}^{n} \left(-\frac{x_i(y_i - x_i w_1)^2}{2\exp(x_i w_2)} + \frac{x_i}{2} \right)$

Question 26. [1 MARK]

Let everything be defined as in the previous question. You want to solve for $\mathbf{w}_{\text{MLE}}$ using gradient descent. Using the partial derivatives you calculated in the previous questions, what would the gradient update rule look like with a constant step size η?

a. $w_1 \leftarrow w_1 - \eta \sum_{i=1}^{n} \left(\frac{x_i(y_i - x_i w_1)}{\exp(x_i w_2)} \right)$,

$w_2 \leftarrow w_2 - \eta \sum_{i=1}^{n} \left(\frac{x_i(y_i - x_i w_1)^2}{2\exp(x_i w_2)} - \frac{x_i}{2} \right)$

b. $w_1 \leftarrow w_1 + \eta \sum_{i=1}^{n} \left(\frac{x_i(y_i - x_i w_1)}{\exp(x_i w_2)} \right)$,

$w_2 \leftarrow w_2 + \eta \sum_{i=1}^{n} \left(\frac{x_i(y_i - x_i w_1)^2}{2\exp(x_i w_2)} - \frac{x_i}{2} \right)$

c. $w_1 \leftarrow w_1 - \eta \sum_{i=1}^{n} \left(\frac{(y_i - x_i w_1)}{2} \right)$,

$w_2 \leftarrow w_2 - \eta \sum_{i=1}^{n} \left(\frac{(y_i - x_i w_1)^2}{2\exp(x_i w_2)} - \frac{x_i}{2} \right)$

d. $w_1 \leftarrow w_1 - \eta \sum_{i=1}^{n} \left(\frac{(y_i - x_i w_1)}{\exp(x_i w_2)} \right)$,

$w_2 \leftarrow w_2 + \eta \sum_{i=1}^{n} \left(\frac{(y_i - x_i w_1)^2}{2} - \frac{x_i}{2} \right)$